

Linear Independent Component Analysis via Optimal Transport

Ashutosh Jha³ · Michel Besserve^{1,2} · Simon Buchholz¹

¹Max Planck Institute for Intelligent Systems, Tübingen ²Institute of Artificial Intelligence, TU Braunschweig ³Methods Center, University of Tübingen

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Code & experiments
github.com/ashutoshjha3103/ot_in_linear_ica

Motivation

Independent Component Analysis (ICA) or blind source separation is a recurring problem across fields: separating CDMA signals in **telecommunications**, isolating neural activity from ocular and muscular artifacts in **EEG/MEG**, identifying information shares across markets in **econometric price discovery**, and comparable source separation tasks in imaging and sensor arrays. ICA is the classical tool for these tasks.

ICA recovers mutually independent source signals $\mathbf{Z}$ from a linear mixture $\mathbf{X} = \mathbf{A}\mathbf{Z}$. If at most one source is Gaussian, $\mathbf{A}$ is identifiable up to permutation and scaling (Comon, 1994).

By the Central Limit Theorem, a mixture of samples from independent non-Gaussian sources is *more* Gaussian than the sources themselves. Recovering an independent source therefore means finding the projection that is *least* Gaussian; maximizing non-Gaussianity is the classical ICA objective. Information theory links non-Gaussianity directly to independence once the data is whitened:

Theorem: Independence & Non-Gaussianity (Cardoso, 2022)

Let $Y \in \mathbb{R}^d$ be whitened ($\text{Cov}(Y) = \mathbf{I}$). The mutual information $\mathcal{I}(Y)$ satisfies:

$$\mathcal{I}(Y) = G(Y) - \sum_i G(Y_i)$$

where $G(\cdot)$ is non-Gaussianity (minimal KL divergence to a Gaussian). Two Pythagorean identities give this: a *product-manifold* identity $D_{\text{KL}}(P_Y \| Q) = \mathcal{I}(Y) + \sum_i D_{\text{KL}}(P_{Y_i} \| q_i)$ and a *Gaussian-manifold* identity $D_{\text{KL}}(P_Y \| \mathcal{N}(\Sigma)) = G(Y) + D_{\text{KL}}(\mathcal{N}(\text{Cov } Y) \| \mathcal{N}(\Sigma))$.

Since $G(Y)$ is rotation-invariant, minimizing dependence is equivalent to **maximizing total marginal non-Gaussianity**.

After whitening, every candidate unmixing direction lies on the orthogonal group $O(d) = \{\mathbf{B} \in \mathbb{R}^{d \times d} : \mathbf{B}\mathbf{B}^\top = \mathbf{I}\}$, the rotations of the whitened data.

Classical proxy contrasts

Because the marginal density needed for exact non-Gaussianity is unknown, classical algorithms substitute a proxy contrast, each with a specific, documented limitation:

FastICA · **logcosh negentropy approximation**

Can hit zero gradient or a singular Hessian on specific non-Gaussian distributions engineered to satisfy the zero-negentropy / vanishing-curvature conditions, stalling the fixed-point solver.

JADE · **fourth-order cumulants**

Reliable cumulant-tensor estimation needs $\mathcal{O}(d^4)$ samples: at $d=30$ that is approximately 810k samples, much more than other ICA methods.

InfoMax · **parametric log-likelihood**

The fixed logistic nonlinearity implicitly assumes sub-Gaussian sources; the score is misspecified on super-Gaussian components (e.g. Laplace, Student-t).

Picard · **L-BFGS on the InfoMax objective**

Converges faster per iteration but shares InfoMax's tanh nonlinearity.

OT-ICA: Method

We measure non-Gaussianity as the squared Wasserstein-2 distance to a standard Gaussian Γ , the minimum-cost coupling between two distributions:

$$W_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|^2 d\pi(x, y)$$

Unlike proxy contrasts, W_2^2 is a true metric, non-zero on every non-Gaussian distribution. Below is OT-ICA's main optimization objective: maximize the contrast over a whitened projection $\mathbf{b}^\top \tilde{\mathbf{X}}$ to find the best unmixing direction:

$$\hat{\mathbf{b}} = \arg \max_{\|\mathbf{b}\|=1} W_2^2(\mathbf{b}^\top \tilde{\mathbf{X}}, \Gamma)$$

Theorem: W_2^2 ICA Contrast

Let $\mathbf{Z} = (Z_1, \dots, Z_d)$ be centered, unit-variance, independent sources with at most one Gaussian and smooth positive densities. For any unit vector α with ≥ 2 nonzero entries:

$$W_2^2(\alpha \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^d \alpha_i^2 W_2^2(Z_i, \Gamma)$$

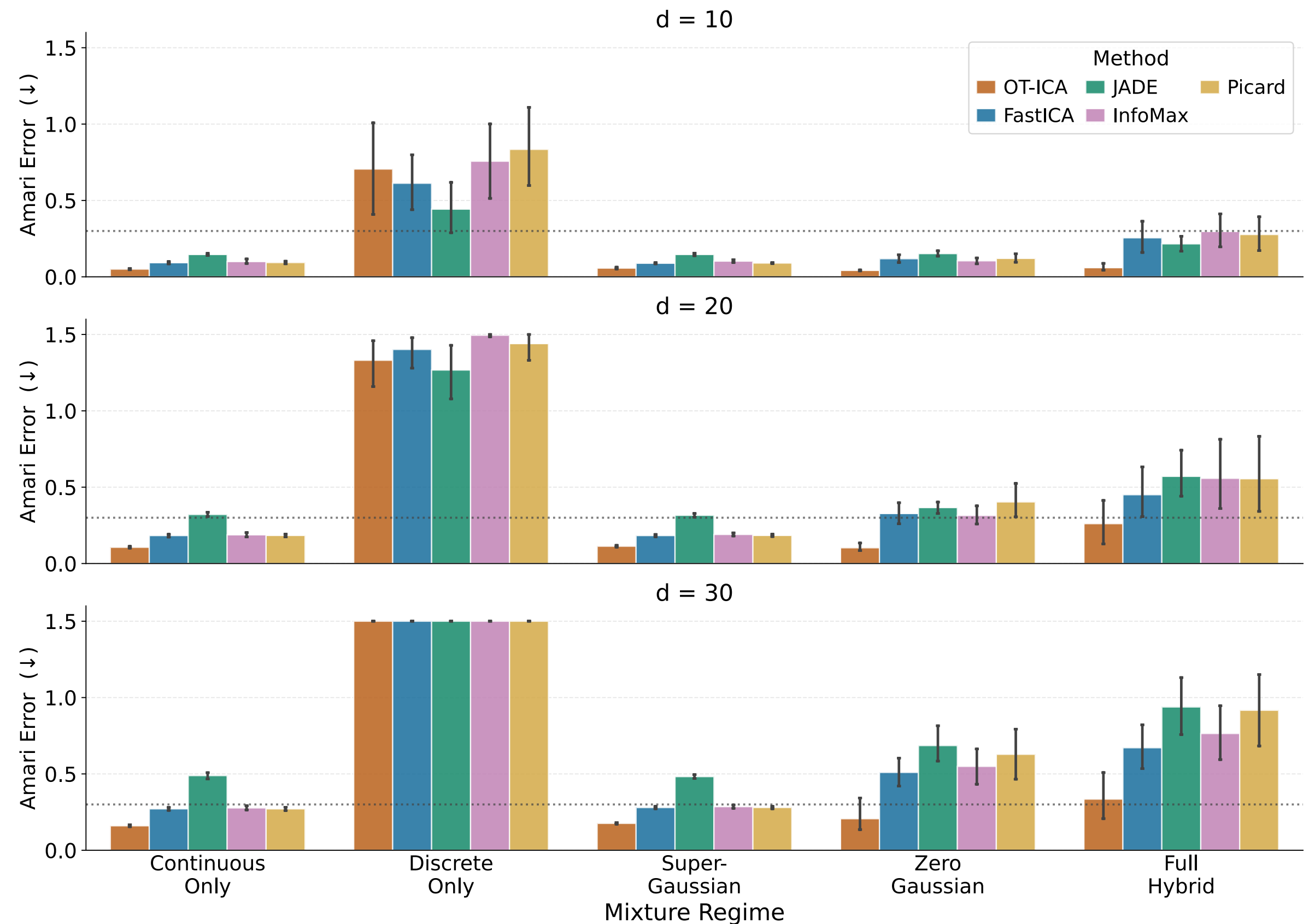
$\Rightarrow$ the contrast is **maximized only at pure independent components**. This extends to joint optimization: for $\mathbf{B} \in O(d)$, the total contrast $\sum_i W_2^2(\mathbf{b}_i^\top \mathbf{Z}, \Gamma)$ is maximal iff $\mathbf{B}$ is a signed permutation.

OT-ICA Algorithm

- Center & whiten: $\tilde{\mathbf{X}} = (\mathbf{X}\mathbf{X}^\top)^{-1/2}\mathbf{X}$; initialize $\mathbf{B} \in O(d)$ randomly, orthogonalize.
- Project a mini-batch: $\mathbf{Y} = \mathbf{B}\tilde{\mathbf{X}}_{\text{batch}}$.
- Gaussian dithering** (introduced here to smooth the empirical CDF before sorting): $\mathbf{Y}_{\text{dither}} = \mathbf{Y} + \mathcal{N}(0, \sigma_{\text{dither}}^2)$.
- Sort each row of $\mathbf{Y}_{\text{dither}}$ to get empirical quantiles; in 1D the optimal coupling is just the sorted match (Brenier's theorem), so no density estimation is needed.
- Loss against analytical Gaussian targets $\mathbf{T}$ (closed-form per-bin expectations, avoiding sampling noise): $\mathcal{L} = \frac{1}{N} \sum_{i=1}^d \|\mathbf{Y}_{\text{sorted}}^{(i)} - \mathbf{T}\|_2^2$.
- Riemannian gradient on $O(d)$: $\nabla_{O(d)} \mathbf{B} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top)\mathbf{B}$, with $\mathbf{G} = \nabla_{\mathbf{B}} \mathcal{L}$; step $\mathbf{B}_{\text{step}} = \mathbf{B} + \eta \nabla_{O(d)} \mathbf{B}$.
- Retract to $O(d)$ by symmetric decorrelation: $\mathbf{B} \leftarrow (\mathbf{B}_{\text{step}}\mathbf{B}_{\text{step}}^\top)^{-1/2}\mathbf{B}_{\text{step}}$.
- Repeat until convergence.

Per-iteration cost: $\mathcal{O}(K(dN \log N + d^3))$ for K restarts, N samples, d dimensions; sorting dominates, and symmetric decorrelation costs $\mathcal{O}(d^3)$.

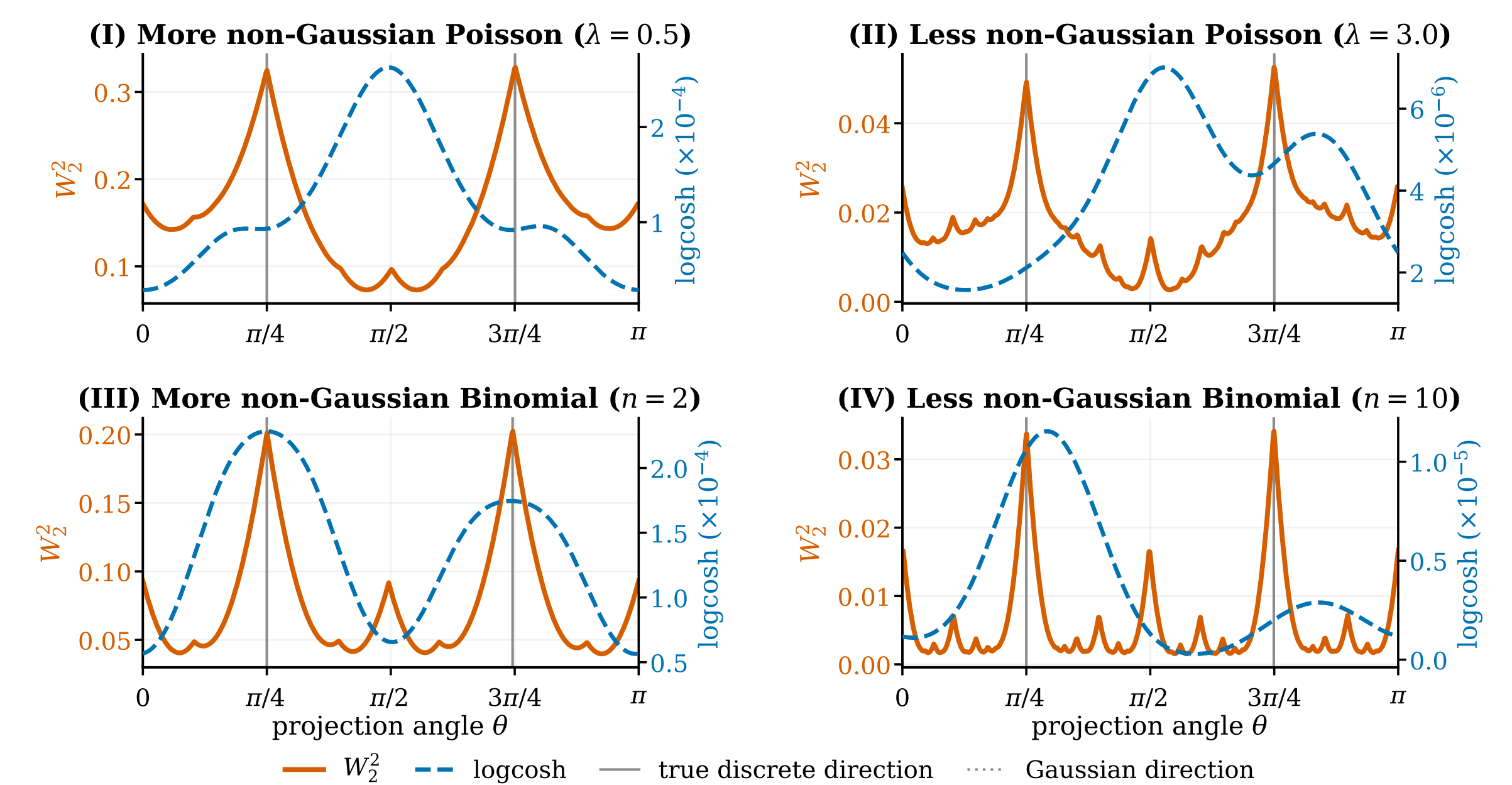
Results



Mean Amari error (1 is better, clipped at 1.5) across five mixture regimes and $d \in \{10, 20, 30\}$ ($N=10,000$, 10 trials, 95% CI). **OT-ICA attains the lowest errors on every continuous and heterogeneous regime**, by 2–4× on Full Hybrid (Continuous, Discrete sources and one Gaussian) mixtures.

Discrete Sources: Contrast vs. Optimization

Landscape on whitened data: two discrete sources ($d = 2$, $N = 10,000$)



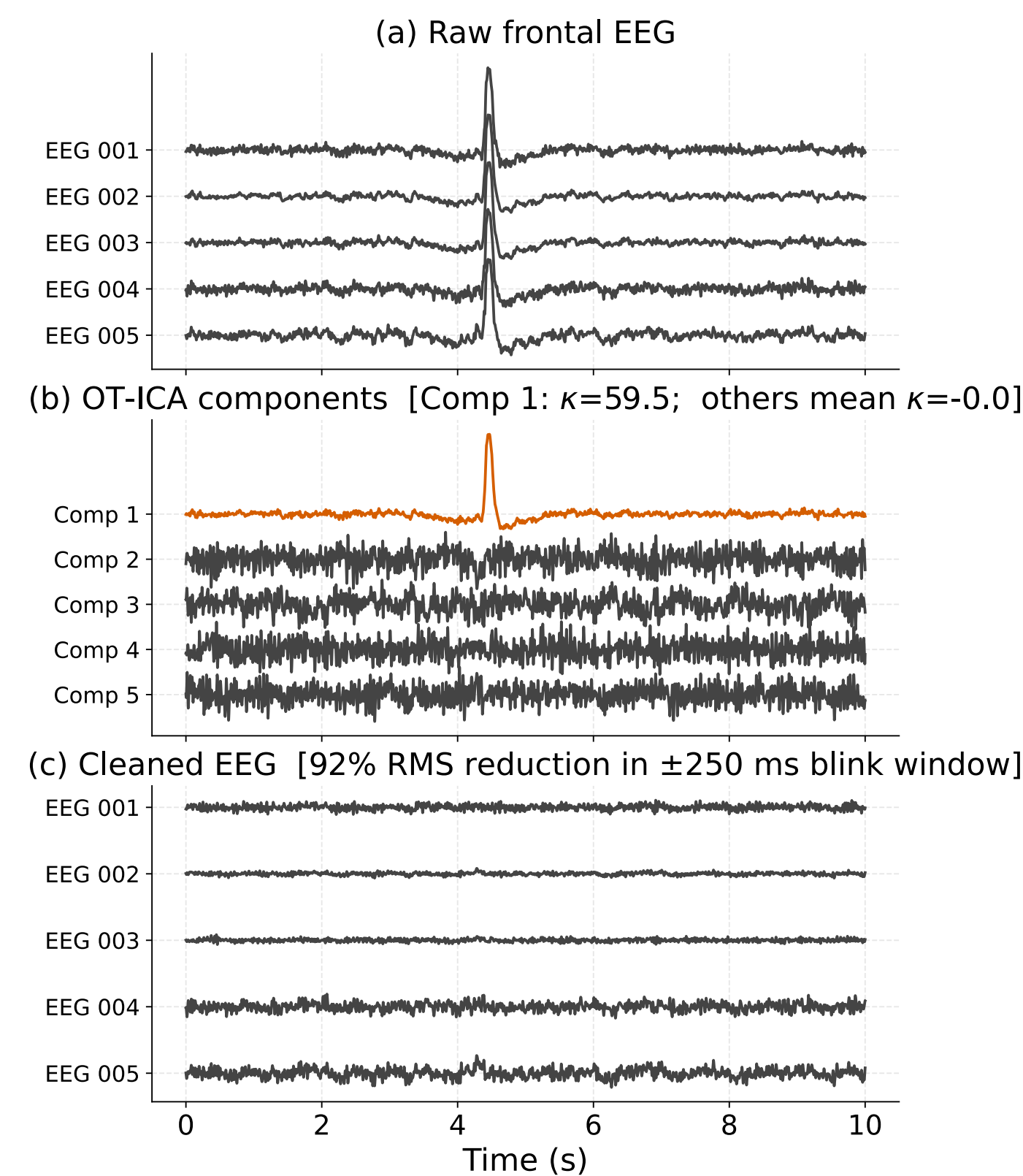
W_2^2 vs. logcosh contrast against projection angle for 2D discrete mixtures. W_2^2 **peaks sharply at the true source directions in every configuration**; logcosh is displaced except in the one case on the bottom left. The contrast signal is intact: it retains $>10\times$ the resolution of logcosh on count data (e.g. Poisson($\lambda=3$): 0.051 vs. ≈ 0). The *Discrete Only* failure at higher d is an **optimization-landscape** problem: step-function CDFs create gradient plateaus that stall the Riemannian solver; it is not a contrast deficiency.

While W_2^2 peaks at the correct locations even in this hard discrete case, the landscape also shows local maxima away from the peak. In future work, we plan to investigate if this is what makes optimization harder in high dimensions, consistent with OT-ICA's weaker performance on high dimensional *Discrete Only* mixtures in the Results above, even though W_2^2 retains a clear contrast signal, unlike the logcosh proxy behind FastICA, which fails to separate sources even in this simple two-source case. If so, the discrete-case bottleneck is the optimization landscape (local maxima and basin collapse), not the contrast itself.

Application

EEG Artifact Removal

OT-ICA on five frontal EEG channels (MNE sample dataset). Blink artifacts are impulsive and super-Gaussian, letting artifact removal proceed without a density model:



- (a) Raw:** five frontal channels with a large blink artifact around $t \approx 4-5$ s.
- (b) OT-ICA components:** Comp 1 isolates the blink (excess kurtosis $\kappa = 59.5$); the remaining four components have near-zero kurtosis (mean $\kappa \approx -0.0$).
- (c) Cleaned:** signal after zeroing Comp 1 and inverting the unmixing matrix, giving a 92% Root Mean Square (RMS) reduction in the ± 250 ms blink window, leaving the signal suitable for further analysis.

1 A Distribution-Free Contrast W_2^2 : replaces proxy negentropy approximations and provably peaks only at independent components.

2 Better Performance: OT-ICA beats FastICA, JADE, InfoMax, and Picard in high dimensions and on more varied, heterogeneous mixtures, as shown in the main results figure, by 2–4× on the hardest cases.

3 Works on Real Data: validated on EEG artifact removal without imposing a distributional model.

Future Work

- On discrete sources, the contrast retains signal but the optimization landscape stalls the solver as d grows.
- Investigating whether basin collapse in the optimization landscape drives this discrete-case failure at higher dimensions.
- Motivates Sinkhorn distances and amortized transport for the discrete case.
- Extending the distribution-free contrast to causal discovery.

References

- Comon. Independent component analysis, a new concept? *Signal Processing*, 1994.
- Hyvärinen & Oja. Independent Component Analysis. Wiley, 2001.
- Cardoso & Souloumiac. Blind beamforming for non-Gaussian signals (JADE). *IEEE Proc. F*, 1993.
- Bell & Sejnowski. An information-maximization approach to blind separation (InfoMax). *Neural Comp.*, 1995.
- Ablin, Cardoso & Gramfort. Faster ICA under orthogonal constraint (Picard). *ICASSP*, 2018.
- Cardoso. Information geometry of ICA, 2022.
- Johnson & Samworth. Central limit theorem and convolution for the Wasserstein metric, 2005.

Contact

Ashutosh Jha
Methods Center, University of Tübingen
ashutoshjha3103@gmail.com